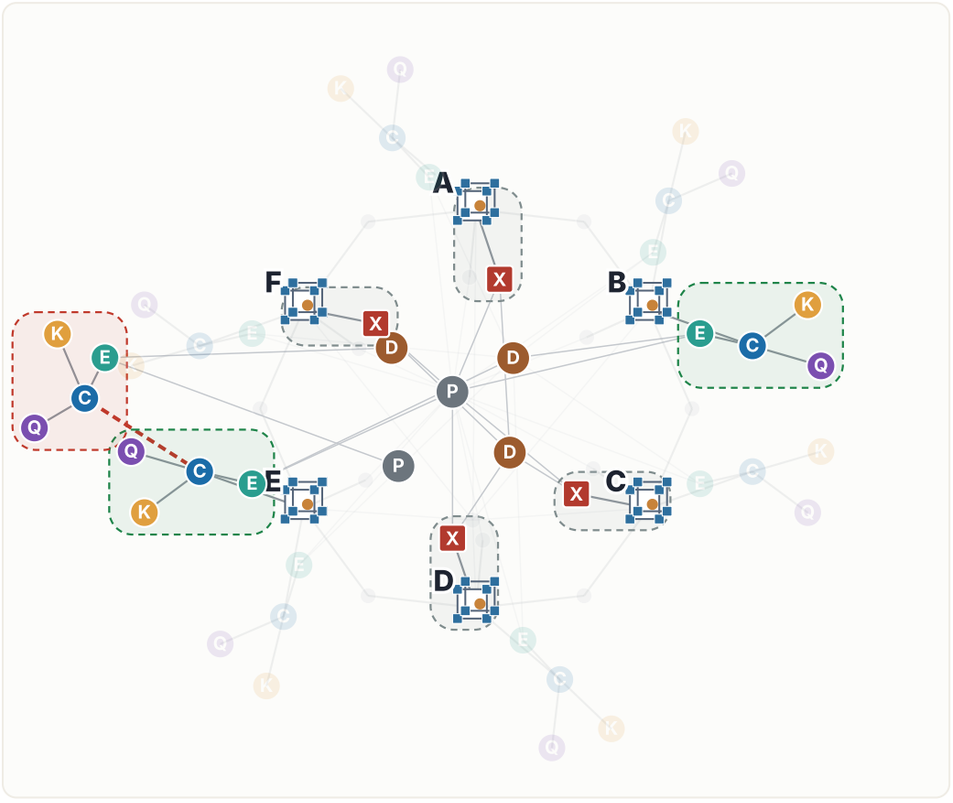
Question: "Which of MOFs A–F has the largest HSE06 band gap?" · **|N(q)|** = 64 nodes, **|J*|** = 27, **k** = 28

MOF	A	B	C	D	E	F
PBE gap (eV)	1.90	1.60	2.20	5.10	2.60	1.20
HSE06 gap in snapshot (eV)	— (exclusion record)	2.85	— (exclusion record)	— (exclusion record)	4.70	— (exclusion record)
ML prediction (HSE06)	—	—	—	—	3.12 (contradicts)	—

a EPS (whole audit units)



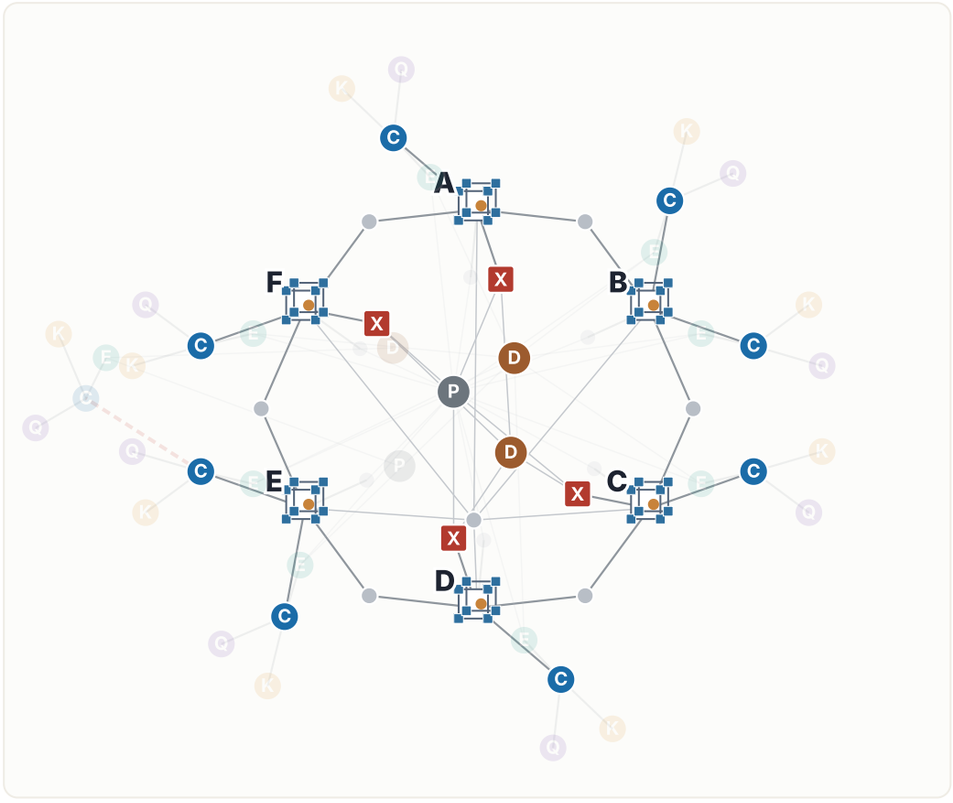
Conservative reader
MOF E: 4.7 eV (HSE06) ⚠ contested

Permissive reader
MOF E: 4.7 eV (HSE06) ⚠ contested

- ✔ Evidence & source (R1)
- ✔ DFT vs. ML (R3)
- ✔ Derivation / replay info (R5)
- ✔ Level of theory (R2)
- ✔ Contradictions (R4)
- ✔ Exclusions (R2x)

27 nodes kept · audit-sufficient: the answer can be replayed from the kept graph alone.

b Personalized PageRank (relevance), same budget



Conservative reader
Abstains (a kept claim has no condition or qualification node)

Permissive reader
MOF D: 5.1 eV (PBE)

- ✖ Evidence & source (R1)
- ✖ DFT vs. ML (R3)
- ✖ Derivation / replay info (R5)
- ✖ Level of theory (R2)
- ✖ Contradictions (R4)
- ✔ Exclusions (R2x)

28 nodes kept · neither correct nor auditable.

material (MOF) · claim · evidence · condition (level of theory) · qualification (DFT / ML) · provenance · derivation · exclusion record · context · contradiction · dashed boxes = audit units kept whole (support · contradiction · exclusion)